THE MINIMUM NUMBER OF ARC-DISJOINT TRANSITIVE TRIPLES IN A TOURNAMENT: A FIRST CERTIFIED DETERMINATION OF $\nu_3(9) = 9$ AND $\nu_3(10) = 12$

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ABSTRACT. For a tournament T let $\nu_3(T)$ be the maximum number of pairwise arc-disjoint transitive triples in T, and let $\nu_3(n) = \min_T \nu_3(T)$ over all n-vertex tournaments. Yuster (2004) conjectured $\nu_3(n) = \lceil n(n-1)/6 - n/3 \rceil$ and verified this for all $n \le 8$ ($n = 8$ by computer, the smaller cases by direct argument), without published certificates; no determination of $\nu_3(9)$ or $\nu_3(10)$, certified or otherwise, appears in the record since. We supply the first certified determination: $\nu_3(9) = 9$ and $\nu_3(10) = 12$, the conjectured values.

The new content is the two lower bounds. An exact branch-and-bound packer was run over every tournament isomorphism class — all 191,536 at $n = 9$ and all 9,733,056 at $n = 10$, enumerated by `gentourng` with counts validated against OEIS A000568 and independently recomputed Burnside numbers — and emitted, for every class, an explicit packing of 9 (resp. 12) arc-disjoint transitive triples, each re-verified from the raw arc data by a standard-library checker sharing no code with the search. The matching upper bounds are not new: they are Yuster's (2004), via oriented Turán blow-ups, restated with a fractional strengthening by Kabiya and Yuster (2008), §2.4. What we add on that side is an artifact: for an explicit minimizer in each order — $C_3[C_3, C_3, C_3]$ at $n = 9$ and $C_3[TT_4, C_3, C_3]$ at $n = 10$ — an LRAT-certified proof that no packing of 10 (resp. 13) triples exists, replayed by an independent checker that regenerates the CNF from its specification.

One trust assumption is declared rather than checked: that `gentourng` (nauty) enumerates the isomorphism classes completely. It was cross-validated exhaustively for $n \le 7$ and count-validated at $n = 9, 10$. Everything else was verified twice: by stdlib-only verifiers, and by an adversarial referee who re-ran the enumeration to byte-level set equality, re-solved both minimizers with an independent exact solver, and exercised both verifiers with negative controls. The next open case, $\nu_3(11)$ (conjectured value 15), is a factor of roughly 93 larger and is within reach of the same pipeline (CPU-days, not CPU-years); it has not been started.

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Computation and authorship. All searches, encodings, and certificate designs in this work were produced by **Claude Fable 5** (Anthropic), directed by the author, on a single Apple M4 laptop. The independent verifiers import only the Python standard library and share no code with the search pipeline; the adversarial referee of §3.2 was a separate agent instance given no access to the pipeline and wrote its own code throughout. This is a factual methods statement, and it is part of the point of the note: the artifacts are designed so that the provenance of the *search* is irrelevant to the validity of the *result*. External tools used by the pipeline only: `gentourng` (nauty 2.9.3), CaDiCaL 3.0.1 (`--lrat --no-binary`), Apple clang 17.0.0, Python 3.

Prior-art record. The Kabiya–Yuster text [KY08] was read in full on August 5, 2026, from the copy saved with the artifacts (an author preprint; section references follow its numbering). The concluding-remarks account in [Yus04] of the verified range and of the upper-bound construction was checked verbatim against the arXiv text (`math/0304180`) the same day; the results of [Yus04] are otherwise used as reported in [KY08]. The novelty sweep of the same date — the Open Problem Garden entry fetched live, the arXiv API, and general web sweeps — found no determination of the integral values $\nu_3(9)$ or $\nu_3(10)$ anywhere in the record since 2008. The upper-bound half of both values was never open; see §1.3. *Draft of August 5, 2026.*

1. Introduction

1.1. **The problem.** A tournament is an orientation of the complete graph. A transitive triple (TT_3) is a 3-vertex subtournament with no directed cycle; a TT_3-packing of a tournament T is a set of pairwise arc-disjoint transitive triples of T, and $\nu_3(T)$ is the maximum size of one. Following Kabiya and Yuster [KY08] we write

$$\nu_3(n) \;=\; \min\{\nu_3(T) : T \text{ a tournament on } n \text{ vertices}\},$$

the value of the packing problem on the *worst* n-vertex tournament. Yuster [Yus04] proved that $\nu_3(n) \geq \frac{51}{392}n^2(1-o(1))$, conjectured the exact value, and verified the conjecture for all $n \leq 8$ — $n = 8$ by computer, the smaller cases by direct argument [Yus04, §4]:

Conjecture (Yuster [Yus04]*; Conjecture 1.1 of* [KY08]*).* $\nu_3(n) = \lceil n(n-1)/6 - n/3 \rceil$.

The conjectured sequence begins, from $n = 3$: $0, 1, 2, 3, 5, 7, 9, 12, 15, \ldots$ Kabiya and Yuster [KY08] improved the asymptotic lower bound to $\frac{41}{300}n^2(1-o(1))$, still the best known, by solving the *fractional* relaxation at $r = 10$ by linear programming ($\nu_3^*(10) = 12$) and transferring the bound to the integral problem with a loss. Their paper is the last progress recorded on the problem's Open Problem Garden entry [OPG], which records no progress beyond 2008.

The state of the integral problem at the start of this work was therefore: conjectured value known for all n; verified for $n \leq 8$ by Yuster ($n = 8$ by computer), with no certificates for that range in [Yus04] or anywhere else in the record; at $n = 9$ and $n = 10$, an upper-bound half that was closed in print (see §1.3) and a lower-bound half with nothing in the record. *No certified determination of* $\nu_3(9)$ *or* $\nu_3(10)$ *existed.*

1.2. **Result.**

Theorem 1.1. $\nu_3(9) = 9$ *and* $\nu_3(10) = 12$*. Consequently, together with the verified range* $n \leq 8$ *(*[Yus04]*, reproduced in §2.2), Yuster's conjectured formula* $\lceil n(n-1)/6 - n/3 \rceil$ *holds for all* $n \leq 10$*, and a counterexample, if one exists, has* $n \geq 11$*.*

Every claim in Theorem 1.1 is backed by a machine-checkable artifact: an explicitly verified witness packing for every one of the $191{,}536 + 9{,}733{,}056$ tournament isomorphism classes (the lower bounds), and LRAT unsatisfiability certificates on explicit minimizer tournaments (the upper bounds), all replayable with standard-library Python and shipped with hashes (§6). This extends the verified range of the conjecture from $n \leq 8$ (uncertified) to $n \leq 10$ (certified).

1.3. **Provenance and credit.** The division of labor between this note and the literature is exact, and stating it precisely is a referee-imposed condition on the writing.

The upper bounds are Yuster's. Section 4 of [Yus04] orients the complete 3-partite Turán graph $V_1 \to V_2 \to V_3 \to V_1$, completes the classes arbitrarily, and observes that every transitive triple of the resulting tournament uses at least one within-class arc; hence no TT_3-packing exceeds the number of within-class arcs, which is exactly $\lceil n(n-1)/6 - n/3 \rceil$. Kabiya and Yuster [KY08, §2.4] restate the construction — crediting the tournaments to [Yus04] — and strengthen the bound to the *fractional* packing number. At $n = 9$ with parts $(3, 3, 3)$ this gives $\nu_3(9) \leq 9$; at $n = 10$ with parts $(4, 3, 3)$ it gives $\nu_3(10) \leq 12$. So the upper-bound half of Theorem 1.1 was never open, and nothing about the *values* 9 and 12 being upper bounds is claimed here. What this note adds on the upper side is a certificate: an LRAT proof, for an explicit minimizer in each order, that one more triple does not fit (§2.4).

The lower bounds are the new content. $\nu_3(9) \geq 9$ and $\nu_3(10) \geq 12$ assert something about *every* tournament of the given order, and for these no argument, computation, or certificate existed in the record. They are established here by exhaustive per-isomorphism-class computation with per-class witnesses (§2.3).

The enumeration is nauty's; the counts are classical. Tournament isomorphism classes were generated by `gentourng` from the nauty suite of McKay and Piperno [MP14], and the class counts 191,536 and 9,733,056 are OEIS sequence A000568 [OEIS], recomputed independently during verification via Burnside's lemma (Davis's formula [Dav54]).

1.4. **What is not claimed.** We do not claim the upper bounds, which are [Yus04]'s (with the fractional strengthening of [KY08]). We do not claim any asymptotic improvement; the best general lower bound remains $\frac{41}{300}n^2(1-o(1))$ of [KY08]. We do not claim the values were unexpected: they are exactly the conjectured ones. We do not claim the conjecture; it remains open for $n \geq 11$. And one link in the chain is trusted rather than machine-checked — the completeness of the `gentourng` enumeration — and §4 states exactly what supports that trust.

2. The two halves of the computation

2.1. **Enumeration.** All tournaments were handled up to isomorphism, which suffices because ν_3 is an isomorphism invariant. `gentourng -q 9` produced 191,536 tournaments and `gentourng -q 10` r`/16`, $r = 0, \ldots, 15$, produced 9,733,056 in sixteen slices (per-slice counts are shipped in `n10_chunk_counts.txt` and sum exactly). Both totals equal A000568 [OEIS]. Each tournament is a line of $\binom{n}{2}$ characters, the upper triangle row by row, `1` at pair (i,j), $i<j$, iff the arc is $i \to j$; the output strings were checked to be pairwise distinct (`sort -u`: 191,536 and 9,733,056).

2.2. **The exact packer, validated on the known range.** `tt3pack.c` (136 lines of C) computes $\nu_3(T)$ exactly by branch and bound: it branches on a live uncovered arc with the fewest free triples through it — either some free triple through it joins the packing, or the arc is marked permanently uncovered — and prunes with the bound $|\text{packing}| + \lfloor \text{live free arcs}/3 \rfloor$. With a target $t > 0$ it stops early as soon as a packing of size $\geq t$ is found and prints it; with target 0 it returns the exact maximum with an optimal packing. Before any new claim was attempted, the packer was run in exact mode over the complete enumeration for $n = 4, \ldots, 8$ and returned minima $1, 2, 3, 5, 7$ — reproducing Yuster's verified range [Yus04] exactly (achieving tournaments and optimal packings in `smalln_table.txt`).

2.3. **Lower bounds: a witness for every isomorphism class.** The sweep `gentourng -q 9 | tt3pack 9 9` terminated with every one of the 191,536 classes reporting a packing of ≥ 9 arc-disjoint transitive triples, each line carrying the packing itself (about a second of wall clock; log `n9_sweep.txt.gz`). The sixteen $n = 10$ slices, run as `gentourng -q 10` r`/16 | tt3pack 10 12`, terminated with all 9,733,056 classes reporting a packing of ≥ 12, about 3 s per slice (logs `n10_sweep_0..15.txt.gz`). Since every tournament is isomorphic to a listed one and ν_3 is invariant,

$$\nu_3(9) \geq 9, \qquad \nu_3(10) \geq 12.$$

These 9,924,592 witness lines are the note's entire novel mathematical content, and they are what the verification of §3 re-checks line by line.

2.4. **Upper bounds: certified optimality on explicit minimizers.** Following [Yus04, §4] and [KY08, §2.4], the candidate minimizers are the cyclic blow-ups $C_3[T_1, T_2, T_3]$: three vertex classes with all arcs $V_1 \to V_2 \to V_3 \to V_1$ and arbitrary tournaments inside the classes. The packer, in exact mode, was run on every within-class variant: all 8 variants with parts $(3,3,3)$ at $n = 9$ have $\nu_3(T) = 9$ exactly, and all 48 variants with parts $(4,3,3)$ at $n = 10$ have $\nu_3(T) = 12$ exactly (`cand9.out`, `cand10.out`) — consistent with [Yus04, §4] and [KY08, §2.4], where the upper bound is proved for an arbitrary within-class completion. The representatives fixed for certification are

$$T_9 = C_3[C_3, C_3, C_3], \qquad T_{10} = C_3[TT_4, C_3, C_3]$$

(arc strings in `minimizer9.bits`, `minimizer10.bits`; T_{10} has the transitive 4-tournament in its 4-class). For each, two artifacts are shipped:

- an optimal packing — 9 triples in T_9, 12 in T_{10} (`minimizer9.witness`, `minimizer10.witness`) — so the values are attained;
- a proof that no larger packing exists: a CNF asserting "T contains 10 (resp. 13) pairwise arc-disjoint transitive triples", refuted by CaDiCaL 3.0.1 [BFF+24] with an LRAT proof [CFHKS17]: `min9_ge10.cnf` (2,430 variables, 5,058 clauses; proof `min9_ge10.lrat`, 419,396 bytes) and `min10_ge13.cnf` (5,740 variables, 11,916 clauses; proof `min10_ge13.lrat`, 1,938,465 bytes).

The encoding (full byte-exact specification in the `encode_cnf.py` docstring) has one variable per transitive triple of T, a binary conflict clause for each pair of triples sharing an arc, and a Sinz sequential at-most-K counter [Sin05] on the negated triple variables, $K = m - k$ with m the number of transitive triples, so that at least k triples must be selected. Its correctness splits asymmetrically. That any satisfying assignment yields k arc-disjoint triples is forced clause by clause (conflict clauses; counter soundness). The direction that makes UNSAT meaningful — any packing of size k extends to a satisfying assignment of the counter variables — is the standard completeness property of the sequential-counter encoding [Sin05], and it sits in the trusted base (§4). As positive controls, the same encoding at the attained value is satisfiable, and was solved as such: `min9_ge9.cnf` and `min10_ge12.cnf` are SAT (CaDiCaL exit 10), so unsatisfiability appears exactly at value + 1. Hence $\nu_3(T_9) = 9$ and $\nu_3(T_{10}) = 12$, giving $\nu_3(9) \leq 9$ and $\nu_3(10) \leq 12$ — the bounds already available on paper from [Yus04, §4], now in checkable form. With §2.3, Theorem 1.1 follows.

Remark 2.1 (The fractional parameter). The blow-up bound holds for the fractional packing number ν_3^* as well — this strengthening is Kabiya–Yuster's [KY08, §2.4] — so $\nu_3^*(9) \leq 9$ and $\nu_3^*(10) \leq 12$; combining with $\nu_3^*(n) \geq \nu_3(n)$ and Theorem 1.1 gives $\nu_3^*(9) = 9$ and $\nu_3^*(10) = 12$. The $r = 10$ value agrees with the LP computation reported in [KY08, §3] and provides an independent derivation of it that does not rest on an uncertified LP solve. This is consistent with the observation of [KY08, §2.4] that their Conjecture 1.1 implies $\nu_3^*(n) = \nu_3(n)$ for all n.

3. Verification

3.1. **Independent stdlib-only verifiers.** Two verifiers, sharing no code with the searcher and importing only the Python standard library, re-check everything the pipeline produced.

`verify_sweep.py` re-checks every one of the 9,924,592 sweep lines from the raw arc string alone: each witness triple is transitive in the stated tournament, no arc is used twice, the count meets the target, the line totals match the expected class counts exactly. All 17 witness files pass with zero failures. A below-target line would be flagged loudly as a candidate counterexample; there were none.

`verify_minimizer.py` re-establishes $\nu_3(T_9) = 9$ and $\nu_3(T_{10}) = 12$ from raw artifacts: it re-checks the optimal packing against the bits; *regenerates* the CNF from the encoding specification with independent code and demands equality with the shipped CNF as a clause multiset; and replays the LRAT proof with its own RUP-with-hints checker — no external tool, and RAT steps are refused outright, which incidentally establishes that both proofs are pure RUP. Both certificates verify.

3.2. **The adversarial-referee protocol.** The result was then subjected to an adversarial referee — a separate agent instance with no access to the pipeline, writing entirely fresh code — whose protocol is part of the contribution and is recorded in the program's results ledger of 2026-08-05. The referee re-ran the enumeration from scratch and confirmed byte-level set equality with the shipped sweeps; recomputed the class counts 191,536, 9,733,056 exactly via Burnside's lemma (Davis's formula [Dav54]), independently of both nauty and OEIS; re-verified all 9,924,592 witness lines with its own sweep verifier (zero failures); re-solved both minimizers with its own exact solver, of a different design and not SAT-based, obtaining 9 and 12; re-ran CaDiCaL afresh on the shipped CNFs; and exercised both stdlib verifiers with negative controls — deliberately corrupted artifacts — all of which were correctly rejected. The referee also imposed the credit framing of §1.3 and caught one error in an internal draft (Remark 5.1). On this protocol the result was confirmed on 2026-08-05.

4. The trusted base

What a skeptic must believe, split into what is re-checked by short readable code and what is assumed.

Machine-checked: every witness packing, per isomorphism class, from raw arc data (§3.1); the totals and per-slice counts; pairwise distinctness of the enumerated strings; the optimal packings in both minimizers; the clause-multiset equality of each shipped CNF with an independent regeneration from the specification; and the LRAT replay of both unsatisfiability proofs, in a checker that refuses RAT steps.

Assumed:

(1) *`gentourng` enumerates completely* — one representative of every isomorphism class. This is the one substantive trust assumption, and it is supported rather than proved: exhaustive cross-validation for $n \le 7$; class counts at $n = 9, 10$ equal to OEIS A000568 [OEIS] and to the referee's independent exact Burnside computation [Dav54]; output strings pairwise distinct; and a fresh re-run by the referee, equal to the shipped enumeration as a set of strings byte for byte. Distinct strings with the right count do not by themselves imply completeness (two listed strings could in principle be isomorphic, hiding a missed class), which is why this item is listed here and not above.

(2) *The sequential-counter encoding is complete* [Sin05]: every packing of size k extends to a satisfying assignment of the counter variables. If the counter accidentally over-constrained, UNSAT would mean less than claimed. ([Sin05] states the property; the short conference paper omits the proofs for space.)

(3) *The arc-string convention is read as written.* Searcher and verifiers necessarily share the *convention* (not code) for interpreting `gentourng`'s output format. A consistent misreading on both sides would not be caught by line re-checking; it is caught instead by the $n \le 8$ reproduction of Yuster's values and by the minimizers, whose blow-up structure is known independently of the format.

(4) CPython and the operating system.

The searcher `tt3pack`, the encoder, CaDiCaL, and the shipped CNF files are *not* in the trusted base: every claim they produced was re-derived from raw data or regenerated and re-checked independently.

5. The frontier: $n = 11$

The next open case is $\nu_3(11)$, conjectured value $\lceil 110/6 - 11/3 \rceil = 15$. There are $A000568(11) = 903{,}753{,}248$ isomorphism classes, a factor of about 93 over the $n = 10$ sweep

— CPU-days, not CPU-years, for the pipeline exactly as shipped (`tt3pack` as written handles $n \le 11$; the 55 arc pairs fit one 64-bit mask). The corresponding minimizer candidates are the blow-ups with parts $(4,4,3)$, whose $6+6+3 = 15$ within-class arcs match the conjectured value, so both halves are within reach of the same two-sided method. This computation is the program's next target; it has not been started, and no result at $n = 11$ is claimed here. The fractional side ($\nu_3^*(r)$ for $r = 11$, which would sharpen the 41/300 of [KY08]) is LP-based, separate, and was not touched.

Remark 5.1 (A corrected estimate). An earlier internal draft of the working notes misstated $A000568(11)$ as $\approx 9.04 \times 10^{11}$ and priced the $n = 11$ sweep accordingly ("$\sim$100 TB of logs"). The referee's exact Burnside computation gives 903,753,248 $\approx 9.04 \times 10^8$ — the same mantissa, three orders of magnitude smaller — and the feasibility claim above rests on the corrected value. We record the error because estimates that die in private make the surviving ones less trustworthy.

6. Artifacts

The artifact directory (`problem-2/` in this program's research tree, distributed with this note) contains the complete pipeline, sweeps, certificates, and verifiers: 148 MB of gzipped $n = 10$ sweep logs in sixteen files, 2.3 MB for $n = 9$, and everything else under 2 MB. The certified release is exactly the 32 files pinned by MD5 digest in `MD5SUMS.txt` and reproduced in Table 1: the sweeps, certificates, minimizers, verifiers, and code. The auxiliary files cited in the text — `n10_chunk_counts.txt`, `smalln_table.txt`, `cand9.out`, `cand10.out`, and the archived primary source — ship alongside them, unpinned; the certified chain of Theorem 1.1 runs entirely through the pinned files.

To replay from scratch on any machine with nauty, CaDiCaL, a C compiler and Python 3:

```
cc -O2 -o tt3pack tt3pack.c
gentourng -q 9  | ./tt3pack 9 9      # lower-bound sweep, n=9
gentourng -q 10 | ./tt3pack 10 12    # lower-bound sweep, n=10
python3 encode_cnf.py 9  "$(cat minimizer9.bits)"  10 min9_ge10.cnf
cadical --lrat --no-binary min9_ge10.cnf min9_ge10.lrat  # s UNSATISFIABLE
python3 verify_sweep.py --n 9 --target 9 --expect 191536 n9_sweep.txt.gz
python3 verify_minimizer.py --n 9 --value 9 --bits minimizer9.bits \
    --witness minimizer9.witness --cnf min9_ge10.cnf --lrat min9_ge10.lrat
```

and the same pattern at $n = 10$ with target 12, value 12, and $K = 13$ files. (The sweep commands stream to standard output; redirect to a file. The first field of each sweep line is a per-run line number, which restarts in every slice: when comparing an unsliced $n = 10$ replay against the sixteen shipped slices, compare the remaining fields. Production ran the sixteen-slice form.) The two verifiers need only the Python standard library; re-checking the shipped artifacts requires neither nauty, nor CaDiCaL, nor a compiler, except for the enumeration itself, which is the declared trust assumption of §4. Tool versions used: nauty 2.9.3, CaDiCaL 3.0.1, Apple clang 17.0.0, CPython 3.

Acknowledgments

The problem, the conjectured formula, the verified range $n \le 8$, and the blow-up upper-bound argument used here at both orders are Raphael Yuster's; the fractional method that still holds the best general lower bound is Mohamad Kabiya and Raphael Yuster's. The enumeration rests entirely on Brendan McKay and Adolfo Piperno's nauty. To Carsten Sinz for the cardinality encoding, to Cruz-Filipe, Heule, Hunt, Kaufmann and Schneider-Kamp for

file	MD5	file	MD5
`tt3pack.c`	`46592e2c895dcd88f485f35f408940dc`	`n10_sweep_2.txt.gz`	`f9c71121e500d9594fab895a8fb9d6ba`
`encode_cnf.py`	`c277f53e7343a44c93375159c248f08a`	`n10_sweep_3.txt.gz`	`4e243e0e91582866523d512669c9eba4`
`verify_sweep.py`	`a3726bfbb524e1d5d29bdfa892776fe0`	`n10_sweep_4.txt.gz`	`d3d96b7b0a78432cb8e80dec1366a36a`
`verify_minimizer.py`	`373639d979b8f44f559ca9ea78ef880b`	`n10_sweep_5.txt.gz`	`bbe411ec67ba2e144ddd1fdb13cb1ce4`
`make_candidates.py`	`546aabe9b6d4a28499da6be2c27e3b45`	`n10_sweep_6.txt.gz`	`9fb5d3da724fcf453451e1ce4e639573`
`minimizer9.bits`	`34ed2059bf5e33ff439e040d0306b7a4`	`n10_sweep_7.txt.gz`	`ece12815439e764a813c99a70b20ddb8`
`minimizer10.bits`	`66d67402bb479a6bc831d10d632c980c`	`n10_sweep_8.txt.gz`	`b9fcb0b5939650e5b5a1f1fa51173438`
`minimizer9.witness`	`35a3d4136122454cb41624bd013c083c`	`n10_sweep_9.txt.gz`	`1cd91ac37afb04e120415e5790a34bb6`
`minimizer10.witness`	`e1b9923118dbe2358007edf31da8da1f`	`n10_sweep_10.txt.gz`	`a1e999d85ba88c3dd131fb123702b46b`
`min9_ge10.cnf`	`f0080051d89ec538e86501efeeaecadf`	`n10_sweep_11.txt.gz`	`64afda1fb8d36eebf04d70a53d2664b1`
`min9_ge10.lrat`	`d13133b094e5bc0ab5fd7f7b7cf1af3b`	`n10_sweep_12.txt.gz`	`90485a8cf9e63e716980c0ef74a6726c`
`min10_ge13.cnf`	`ce16d4fc654198dcb1e3ff0782dd5b1a`	`n10_sweep_13.txt.gz`	`0109f71fbb2a111405b175b6c40f2120`
`min10_ge13.lrat`	`ad546eece47db3d3f7abf9baacc859bf`	`n10_sweep_14.txt.gz`	`a1bb606d782c973e784672c6b90a3f11`
`min9_ge9.cnf`	`a9acac4e212214975eb28f3690d21f86`	`n10_sweep_15.txt.gz`	`a91a473a46034db7d448cd2eb6165019`
`min10_ge12.cnf`	`eb5ae68eb5262236e2c0c8e4947b9c25`	`n9_sweep.txt.gz`	`b43c323b55c2f199d94cacc025d3d15b`
`n10_sweep_0.txt.gz`	`5069371e1500dda7eabdf5fe7a0d4e35`		
`n10_sweep_1.txt.gz`	`73e36c3db19e0815e57de78761d92dd2`		

TABLE 1. MD5 digests of the released artifacts (the 32 entries of `MD5SUMS.txt`).

LRAT, and to Armin Biere and coauthors for CaDiCaL. The computation and drafting were AI-assisted as stated in the first footnote; the adversarial referee's insistence on the credit framing of §1.3 made this a better and a more honest note.

References

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[MP14] B. D. McKay and A. Piperno, *Practical graph isomorphism, II*, J. Symbolic Comput. **60** (2014), 94–112. Version used here: nauty 2.9.3 (`gentourng`).

[Dav54] R. L. Davis, *Structures of dominance relations*, Bull. Math. Biophys. **16** (1954), 131–140.

[OEIS] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, sequence A000568 (the number of tournaments on n unlabeled nodes), `https://oeis.org/A000568`.

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